Assignment 2

Study group: ANDERS WHEELMAN, GUSTAV HELMET AND SIGRID SNAPFIELD.
Due 28/9

Assignment 2 part 1: Basic description of language development

Study group: Anders, Gustav H & Sigrid
Due 16/9
Link to code: https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2
The file is called 'FINAL_A2_P1_LanguageASD_instructions.Rmd'

1. **Describe your sample (n, age, gender, clinical and cognitive features of the two groups) and critically assess whether the groups (ASD and TD) are balanced**

Descriptive statistics for the overall sample

(Anders) The groups are relatively balanced according to ethnicity (the study was executed in Denmark), non-verbal IQ and number of children in each group. The participants were, however, mostly male.
Overall assessment of the data:

- n = 372 (ASD = 176, TD = 196)
- men > women in both groups (ADS: 150 M, 26 F, TD: 160 M, 36 F)
- ethnicity = white > other ethnicities (ASD: 130 white kids, other minorities: 46, TD: 190 white kids, other minorities: 6)
- Non-verbal IQ: ASD mean 33.15, TD mean 35.8

We see clear differences between ASD and TD when it comes to measures of verbal IQ and ADOS. Across all visits we also see the following difference in their linguistic capacities.

- We see a clear difference in autistic measure, ADOS (mean of ADS = 13.8, mean of TD = 1.2)
- Verbal IQ: ASD mean 22.62, TD mean: 29.29
- Mean MLU for the children: ASD: 1.6, TD: 2.3

- Mean Types (unique words used): ASD: 77, TD: 129
- Mean Tokens (amount of words used): ASD: 293, TD: 475

Descriptive statistics for visit 1

(Gustav) The ASD children were matched with TD children by their linguistic skills. This can be seen by the descriptive statistics of the children's linguistic skills at their first visit. This entails that the age difference between the two groups is approximately 1 year at their first visit.

- Age (in months): mean at first visit ASD: 33, TD: 20.4
- The children had similar mean MLU at first visit: TD: 1.37, ASD: 1.24
- Mean Types at first visit (unique words used): ASD: 42, TD: 45
- Mean Tokens at first visit (amount of words used): ASD: 161, TD: 194

2. Describe linguistic development (in terms of MLU over time) in TD and ASD children (as a function of group).

Motivation for determining predictors

(Sigrid)

Fixed effects: As a measure of time, we have chosen to enter "age" and not "visit" in our model. Taken visit by visit, there is a skew in age since autistic children had a higher age at each visit than typically developing children. Thus, "visit" has a skewed representation of the children's age. Furthermore, the time between each visit differs from visit to visit. Using 'age' as a predictor and not 'visit' accounts for this difference. However the limitation of entering age is that there is a difference in the maximal age for the two groups. TD have a max age of 45 months, whereas ASD have a max age of 62 months.

Random effects: We had random intercepts for each subject and random slopes for age to account for the subjects' individual differences in both their skills at start and their different learning speed.

Eyeballing the data

Eyeballing the data gave the indication that there might be an interaction effect between age and diagnosis, and that the random effects are important especially for the ASD group where participants have very different developments (see figure 1).

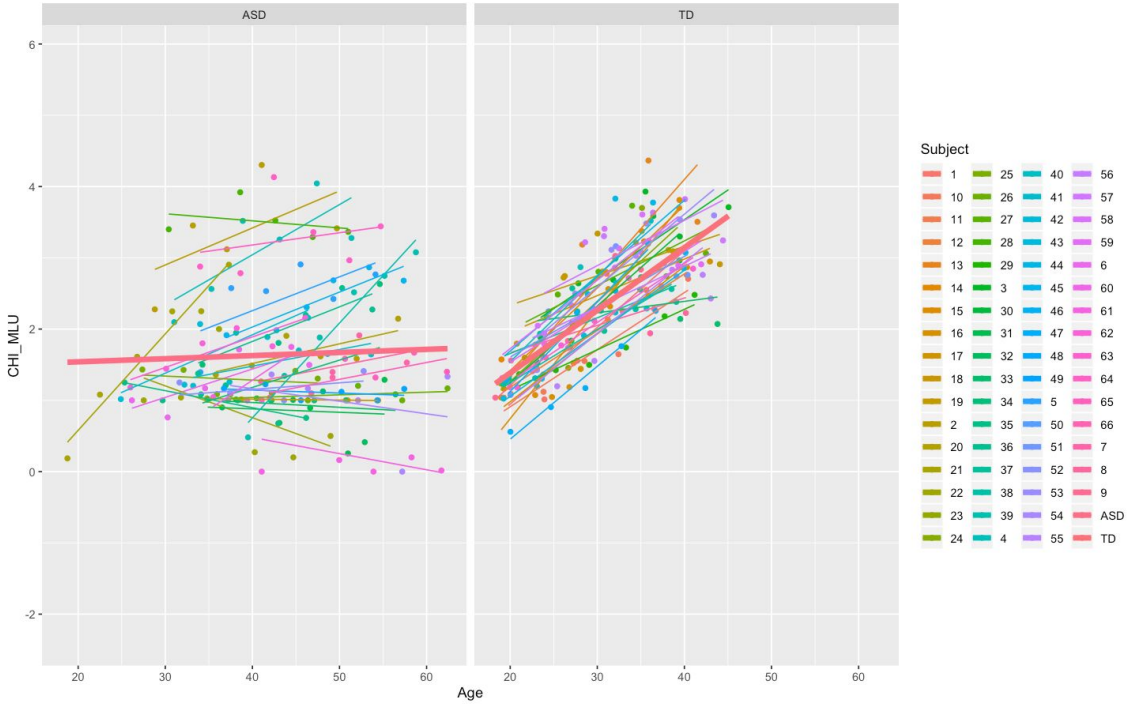

Figure 1: CHI_MLU plotted against age. Random slopes and intercept for each participant.

The model

(Anders) We have built a linear mixed effects growth model using the following syntax and package lme4:

*model = lmer(child_MLU ~ Diagnosis * Age + (Age|Subject) + I(Age^2))*

Results

(Anders) There was a significant interaction effect between diagnosis and age, $\beta = 0.030$, SE = 0.012, $t = 2.542$, $p < .01$.

The interaction effect makes the main effects of diagnosis and age, respectively, useless, since they can only be interpreted when taking the other into account.

Fixed effects alone accounted for roughly 38 % of the variance. Fixed effects and random effects together accounted for roughly 82 % of the variance.

An anova comparing our model with a model excluding the interaction Diagnosis:Age, showed that the interaction made our model significantly better (BIC = 681 and BIC = 612 respectively), thus confirming the importance of the interaction.

Description of results

(Gustav) Children who are typically developing have a significantly higher mean length of utterance as they develop compared to children who have an autistic spectrum disorder. For each month in their

development the mean length/utterance of typical children are 0.003 syllables longer than children with autism spectrum disorder. This is the interaction effect.

Plot of the models predictions

(Gustav) Figure 2 shows the actual values (blue) plotted against the predicted values (red). However these predictions are not sensible, since we didn't split the dataset into training and testing. The predicted values merely show that our model is good at explaining the data.

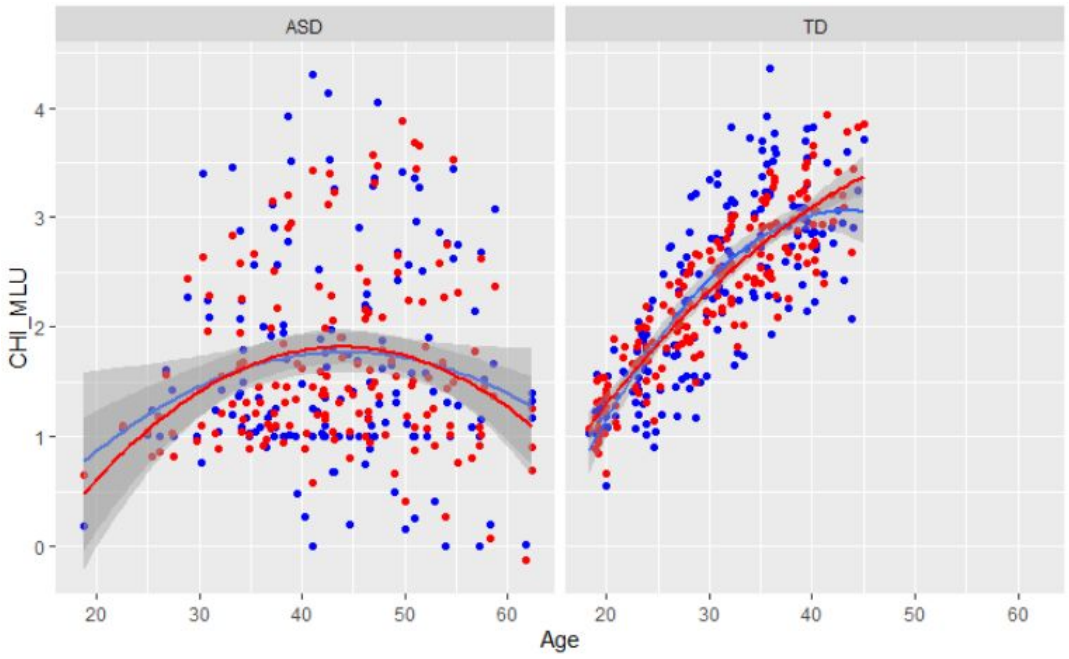

Figure 2: actual values of the mean length of utterance for children (blue) compared to fitted values (red).

3. Describe how parental use of language (in terms of MLU) changes over time. What do you think is going on?

Eyeballing the data

(Sigrid) Eyeballing the data showed that an interaction effect between age and diagnosis probably was not present. It also showed that random effects would not account for as much variance as it did with child mean length utterances.

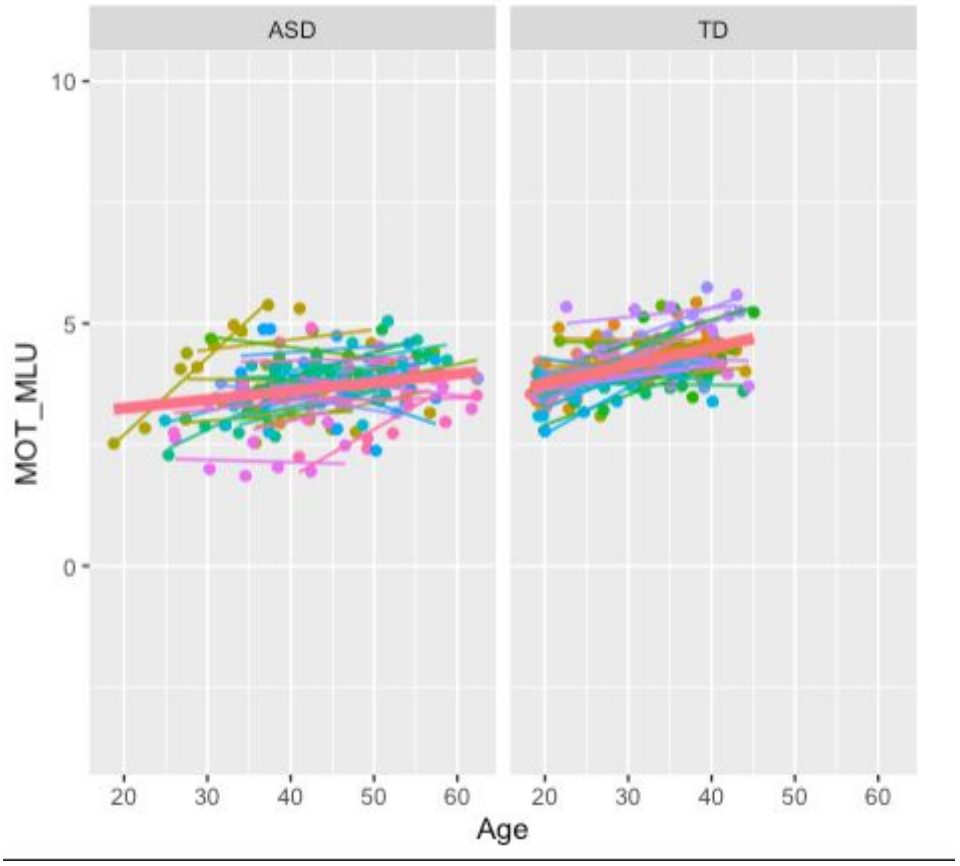

Figure 3: Mother MLU according to the diagnosis and age of the child

Results

(Sigrid) To assess the parental use of language (also in terms of MLU), a model was built using the following syntax:

model = lmer(Mother_MLU ~ Diagnosis + Age + (Age|SUBJ), df)

There was a significant effect of diagnosis; the mother of the child had a lower MLU if the child had the diagnosis of ASD, $\beta = 2.53$, SE = 0.18, $t(100.1) = 14.375$, $p < .0001$, compared to the MLU of mothers with typically developing children, $\beta = 3.314$, SE = 0.12, $t(61.73) = 6.31$, $p < .0001$. Moreover, there was a significant effect of age; the older the child got (in months) the larger MLU of the mother, $\beta = 0.0277$, SE = 0.004, $t(58.95) = 6.865$, $p < .0001$.

Fixed effects alone accounted for roughly 20 % of the variance. Fixed effects and random effects together accounted for roughly 68 % of the variance.

It seems that the mothers of typically developing children (TD) talk more to their children than mothers of children with ASD, and that the MLU of mothers increases as the age of the child increases. This makes sense since TD children have a larger MLU than ASD children.

The results are depicted in figure 4.

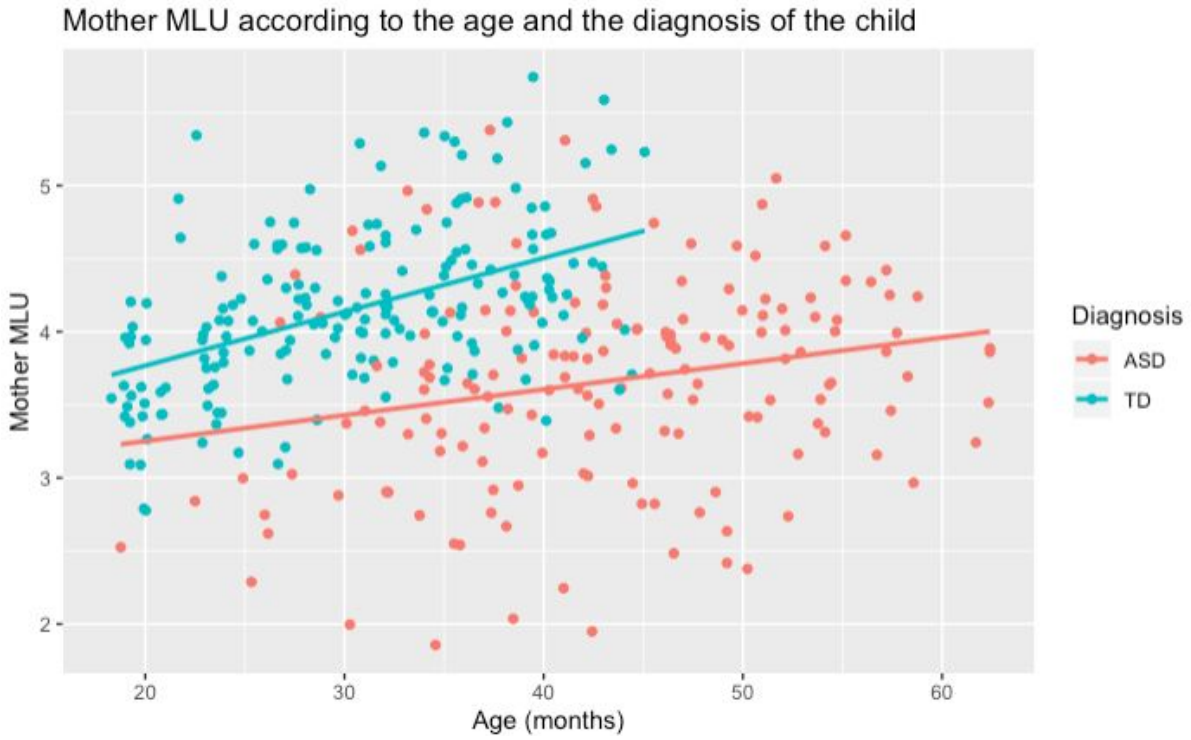

Figure 4: Mother MLU according to the age and diagnosis of the child

4. Include individual differences in your model of language development (in children). Identify the best model.

(Anders) We have conducted a post-hoc exploratory analysis to see the possible effect of other independent variables.
We found that 'ethnicity' was a significant predictor. However, this variable is not very well balanced in each level (e.g. there is only one asian subject), thus, we discard this variable as a valid predictor. 'Gender' was found to be non-significant.

The variables 'non-verbal-IQ', 'verbal-IQ' and 'Socialization' were all found to be significant. These results can be used to make new hypotheses in a follow up study. However, since we did not include them in our hypothesis we can't use them in the current analysis.

Assignment 2 part 2

Expmeth 3, Sigrid Snapfield, Gustav Helmet, Anders Seabridge Wheelman
Due 23/9 2020

The code can be found here:
https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2_2

1. Discuss the differences in performance of your model in training and testing data

(Gustav) The model was built using the following syntax:
m8 <- CHI_MLU ~ Diagnosis * Age + (Age|Subject) + I(Age^2)

Testing the performance of the model on the data it was trained on vs test data

(Gustav)
Performance of model on the training data:
RMSE was calculated for fitted values to the model using the function fitted(). The training data was found to have an RMSE of 0.34 when compared to fitted values.

Performance of model on the test data:
RMSE was calculated for predicted values of the model using the function predict(). The training data was found to have an RMSE of 1.35 when compared to predicted values.

Thus, the model was performing better in the training data than in the test data. This makes sense since the model was created from the training data, meaning that the model is the best fit possible for the training data. Therefore it makes sense theoretically that the performance of the model on new data is not as good as its performance on its training data.

Cross validation results of model m8

(Sigrid) When cross validating the training data using 10 folds, we get a training RMSE value of 0.3524 and a test RMSE value of 0.6765. The RMSE scores of the model after it has been

cross-validated are approximately half the size of the RMSE scores for the model that has not been cross validated several times. These results suggest that cross validation gives less prediction error and is thereby a better method for testing models.

2. Which individual differences should be included in a model that maximizes your ability to explain/predict new data?

(Anders) In addition to the model mentioned above (m8), we created 7 other models:

```
m1 <-  CHI_MLU ~ Diagnosis + Age + (1|Subject)

m2 <-  CHI_MLU ~ Diagnosis + Age + (Age|Subject)

m3 <- CHI_MLU ~ Diagnosis + Age + (1|Subject)+ I(Age^2)

m4 <- CHI_MLU ~ Diagnosis + Age + (Age|Subject)+ I(Age^2)

m5 <- CHI_MLU ~ Diagnosis * Age + (1|Subject)

m6 <- CHI_MLU ~ Diagnosis * Age + (Age|Subject)

m7 <- CHI_MLU ~ Diagnosis * Age + (1|Subject)+I(Age^2)
```

We made a 10-fold cross-validation to compare the performance of all the models (see table 1). We found that the rmse-value for test-data was best (i.e. lowest) for m8, the fully complex model.

Thus, the most complex model is the best fit to new data which it has not been trained on. Normally, we might think that the most complex model would overfit and not generalize well to new data. However, here we find that it is indeed the most complex model which generalizes the best.

Type	Train	Test
rmse_m1[11]	0.4674175	0.7781820
rmse_m2[11]	0.3505257	0.8391998
rmse_m3[11]	0.3943179	0.6950842
rmse_m4[11]	0.3425541	0.6954798
rmse_m5[11]	0.4100964	0.7132800
rmse_m6[11]	0.3613275	0.6956282
rmse_m7[11]	0.3918088	0.6937753
rmse_m8[11]	0.3509706	0.6972461

Table 1: mean rmse-values for cross-validation of model m1-m8 with 10 folds.

Bonus Question 1: What is the effect of changing the number of folds? Can you plot RMSE as a function of number of folds?

(Gustav) In general, the more folds, the lower the value of RMSE. Thus, the more folds the better (given that the data set is big enough). However, when increasing the number of folds on the training data set, the RMSE value does not seem to decrease. It might be due to outliers (see figure 1).

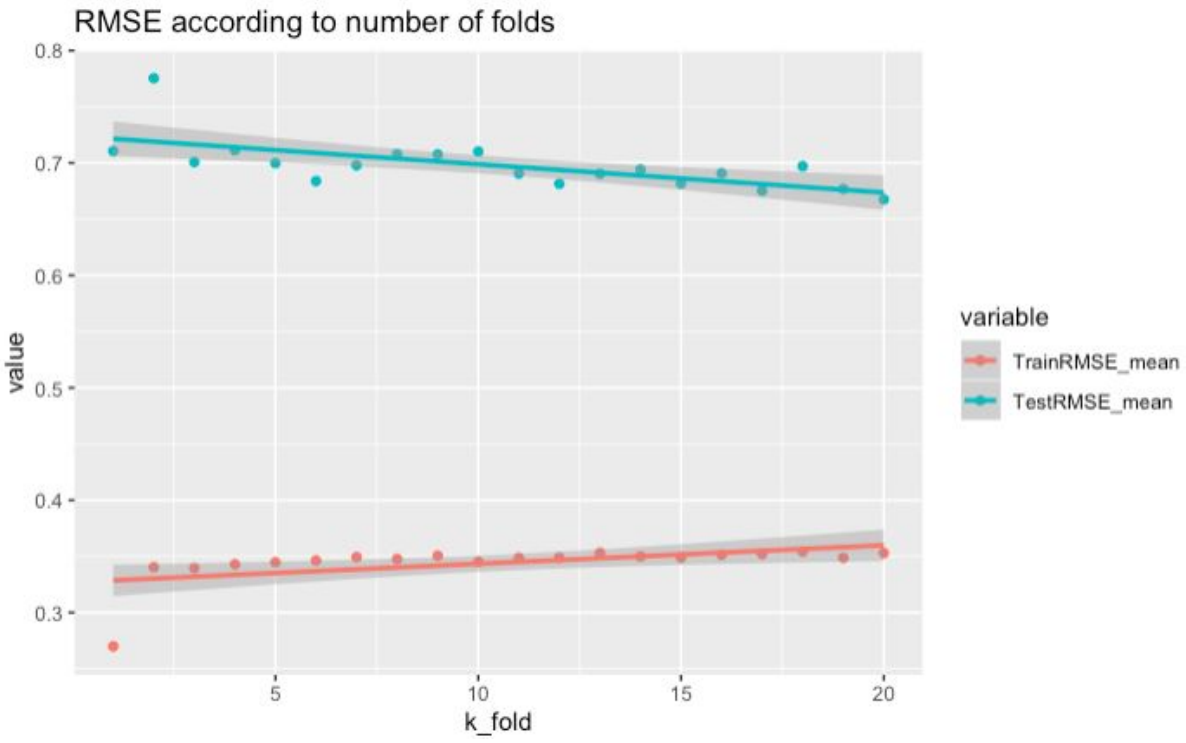

Figure 1: mean rmse-values for training and test data as the number of folds increases.

Bonus Question 2: compare the cross-validated predictive error against the actual predictive error on the test data

(Sigrid) The mean predictive error for the test data was 0.98.
We calculated this value by subtracting the predicted values of m8 by the test data and taking the mean of these values.

The mean predictive error for the cross validated model was 0.79. We calculated the value by extracting the average predictive error for each fold and then getting the mean of the average predictive error for each fold

Thus, the cross-validated predictive error is lower than the actual predictive error of the test data. This confirms that cross validation is a good method for lowering the predictive error.

3. Predict a new kid's performance (let's call him Bernie) and discuss it against expected performance of the two groups

(Anders) The absolute difference between the MLU of Bernie (subject 2) and the MLU of typically developing children at each visit can be found in the column 'Absolute_dif' in table 2.

Subject <dbl>	Diagnosis <chr>	Visit <dbl>	CHI_MLU <dbl>	mean_td <dbl>	mean_asd <dbl>	Age <dbl>	Absolute_dif <dbl>
2	ASD	1	1.984456	1.310174	1.304476	27.80	0.6742815
2	ASD	2	2.544444	1.778062	1.443683	33.33	0.7663823
2	ASD	3	3.353191	2.229037	1.799177	36.23	1.1241544
2	ASD	4	3.183099	2.734681	1.815197	40.03	0.4484175
2	ASD	5	3.173252	2.969893	1.598512	45.47	0.2033593
2	ASD	6	3.448413	2.927875	1.940044	49.30	0.5205373

Table 2: The data of Bernie (subject 2) including the absolute difference of child MLU between Bernie and TD children.

It is clear that the MLU of Bernie is better than both the averaged MLU of typically developing children and that of children with autism spectrum disorder (ASD) (figure 2).

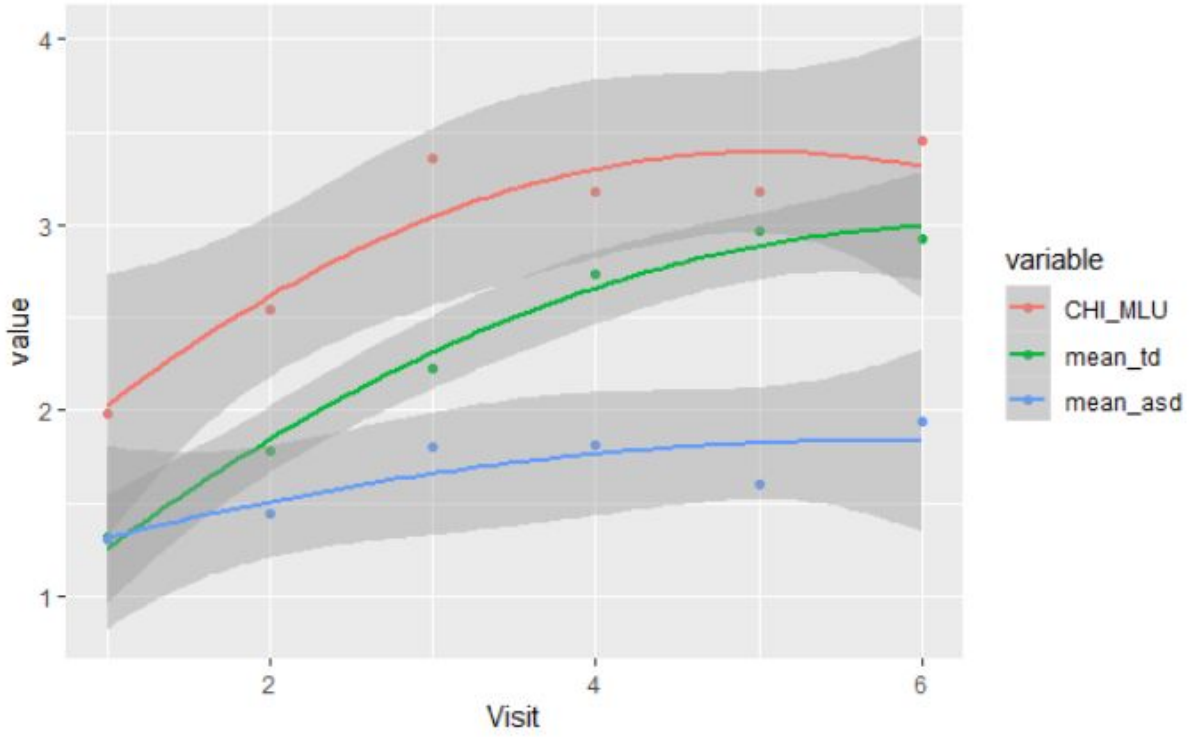

Figure 2: mean CHI_MLU of Bernie is better across visits than mean MLU of typically developing children and autistic children.

(Gustav) When using our model to predict the MLU of Bernie at visit 6, we find that Bernie is doing better than expected. The predicted MLU of Bernie at visit 6 is 1.5988 phonemes, while the actual MLU of Bernie at visit 6 is 3.4484. Thus, Bernie has a MLU which is 1.8496 higher than the predicted MLU of ASD children at visit 6. This is a good example of why it is important to remember that our model is just a model, and there will always be individual differences and thereby subjects who stand out.

Assignment 2 part 3

Expmeth 3, Sigrid Snapfield, Gustav Helmet, Anders Seabridge Wheelman
Due 30/9 2020

The code can be found here:
https://github.com/Study-Group-2-0/Exp_meth_3/tree/master/Assignment_2_3
The file is called 'FOR_GRADING_A2_P3'

In this part of the assignment, we try to figure out how a new study should be planned (i.e. how many participants are needed) in order to have enough power to replicate the findings (ensuring our sample size is adequate, our alpha at 0.05 and our beta at 0.8):

Introduction

(Sigrid) We will be working with two datasets: dataset A, which entails a full dataset consisting of 346 rows (66 participants), and dataset B, which functions as a "pilot" dataset consisting of 36 rows (6 participants). In the preprocessing of the data, 26 rows were excluded in dataset A by criteria of missing values in the variables "CHI_MLU" (child MLU) and "age".

All power analyses were conducted with the following model using the package lme4 (Bates, Maechler, Bolker & Walker, 2015).

model1 = child_MLU ~ Diagnosis * Visit + (Visit|Subject)

From the previous parts of this assignment we know that a significant effect of the interaction between Diagnosis and Visit can be found. Thus, we will only examine the interaction effect and thereby disregard the main effects when conducting the following power analyses.

All power analyses were conducted by running 50 simulations using the package simr (Green & MacLeod, 2016) to test the sufficient number of participants needed to get a power of at least 80% at an alpha level of 0.05.

1. if we trust the estimates of the current study. Report the power analysis and comment on what you can (or cannot) use its estimates for.

(Anders) A power analysis on dataset A using model1 showed that the interaction Diagnosis:Visit had a power of 100 %, CI 95 % [93, 100]. In figure 1 it can be seen that 25 participants are needed to obtain sufficient power of 80%. The current study has 66 participants. We can thereby conclude that the current study was overpowered.

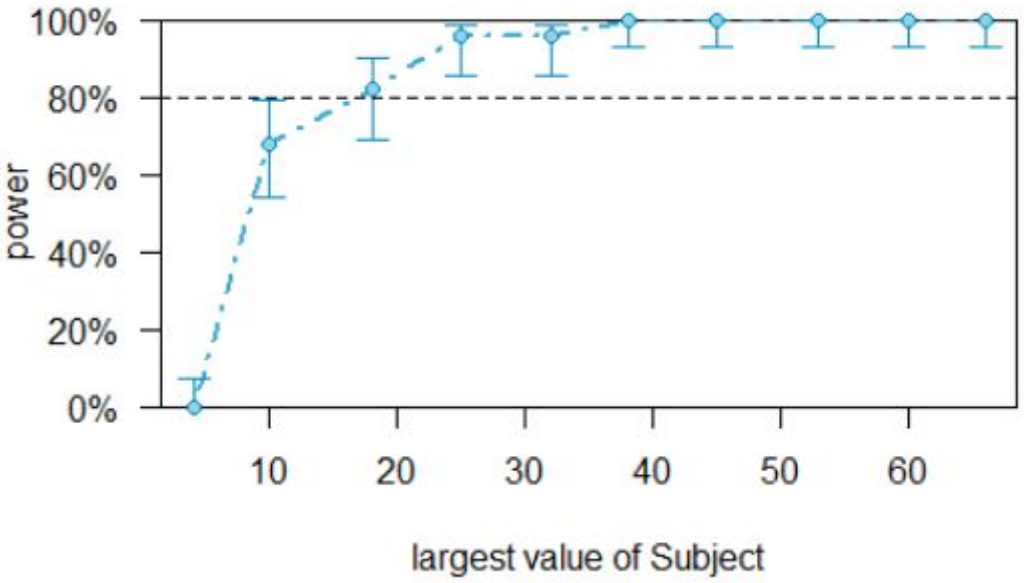

Figure 1: Power plot on dataset A. The effect size of Diagnosis:Visit is 0.25. 25 participants are needed to obtain a power of >80%.

(Gustav) The estimate of Diagnosis:Visit is 0.25. Thus, typically developing children will have a mean length of utterance that is 1.5 syllables longer than children with an autism spectrum disorder at visit 6. We argue that this estimate is too high, and would not necessarily be replicated in a follow-up study. Therefore, we lower the estimate for a more conservative power analysis.

2. if we are skeptical of the current study. Report the power analysis and comment on what you can (or cannot) use its estimates for.

Full study

(Sigrid) In order to run a more conservative power analysis the effect size can be set to a lower value. However, the effect size should still be theoretical justifiable. We have decided that a difference of minimum 1 syllable per utterance from visit 1 to visit 6 is sufficient for a meaningful effect size. Thus, the lowest justifiable effect size would be 1 syllable divided by 6 visits, which equals 0.167.

A power analysis was conducted on model 1 by simulating dataset A at an effect size of 0.167 (see the section 'Introduction' for details) . Results showed that a total sample of 53 participants was required to achieve a power of 94% with CI 95% [83.45, 98.75] (figure 2).

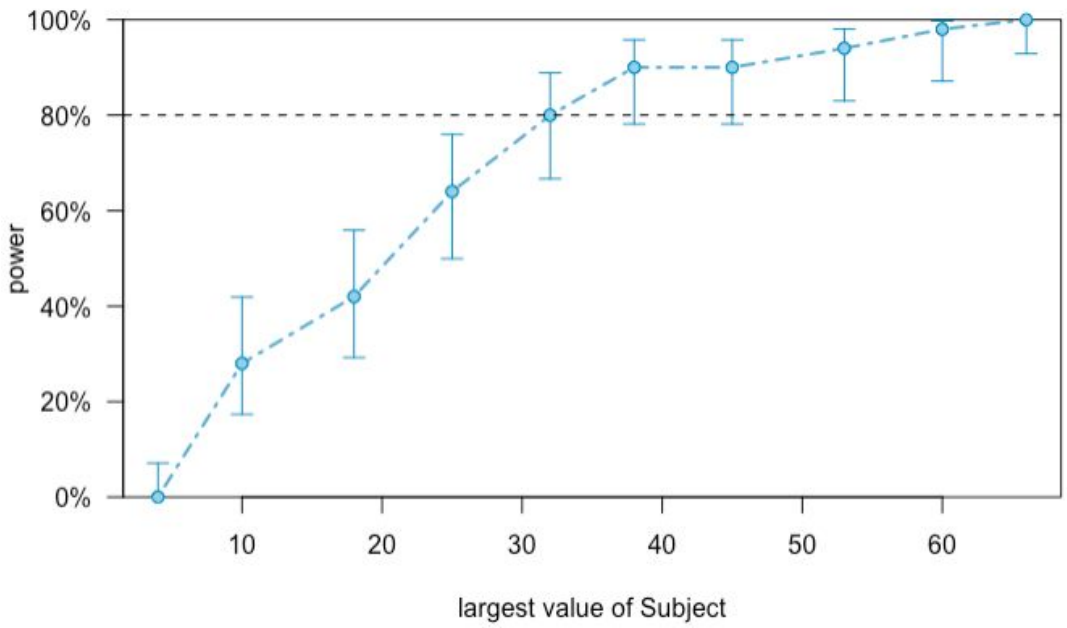

Figure 2: Power curve for an effect size of 0.167 simulating dataset A on model 1. 53 participants are sufficient to get a power of >80%.

Pilot study

(Anders) Since we had access to another dataset (dataset B) we decided to analyse it as it was a pilot study with 6 participants. Note that this is not a legit pilot-study, but merely an exploratory exercise to investigate the validity of a power analysis on a pilot study.

First we conducted a power analysis on dataset B. We found that with 6 participants and an effect size of 0.167 the pilot study had a power of 10 % with CI 95% [3.33, 21.81]. We then conducted a power analysis where the number of participants had been extended, to investigate the sufficient number of participants needed to obtain a power of >80%. We found that a power of 92% with CI 95% [80.77,97.78] were reached with 48 participants (figure 3). This estimate was a good approximation of the estimate of 53 participants in our analysis using dataset A. However, it is important to take into account that it is assumed that the pilot study data is representative of the population, which might be violated since dataset B only had 6 participants, that is n = 3 in each group (ASD, TD).
All in all this experiment supports the notion that extending the number of subjects from a pilot study gives a good approximation of the number of participants needed to obtain a power of >80%.

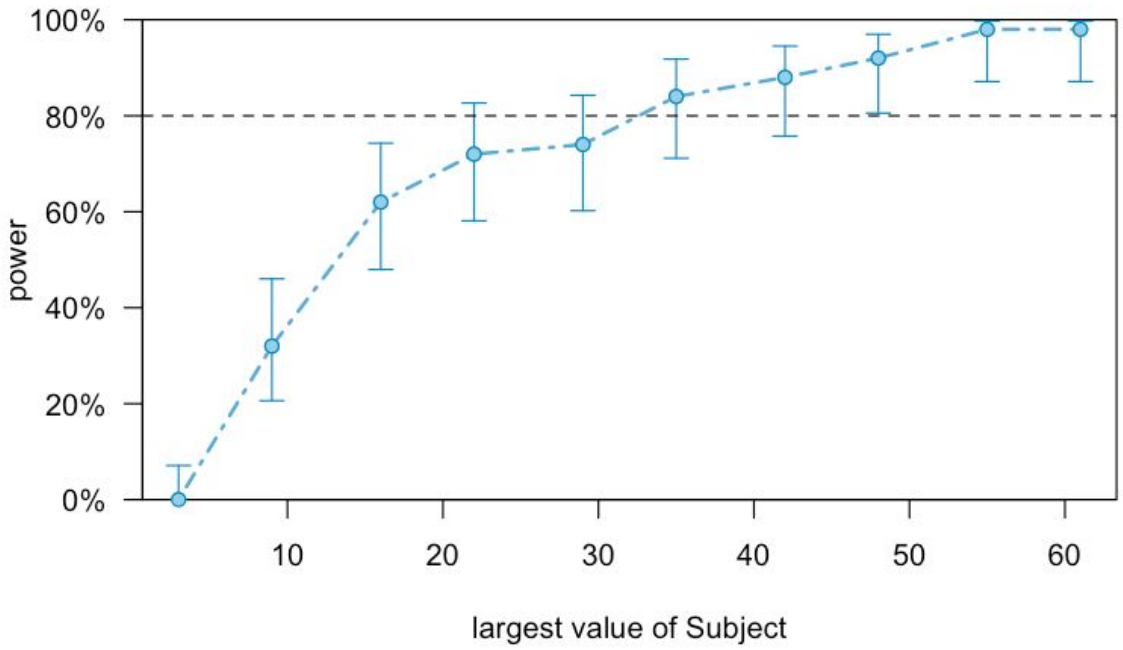

Figure 3: Power curve for an effect size of 0.167 simulating dataset B on model 1 as a pilot study. 48 participants are sufficient to get a power of >80%.

3. if we only have access to 30 participants. Identify the power for each relevant effect and discuss whether it's worth to run the study and why

(Gustav) We assessed the power for the interaction effect between Diagnosis and Visit with n = 30 using the function extend(n=30). We found that with 30 participants and an effect size of 0.167, the study had a power of 86 % with CI 95% [73.26 , 94.18]. The obtained power of 30 participants is insufficient because of the large confidence interval that drops below 80%.
It is therefore not worth it to run the study with only 30 participants.

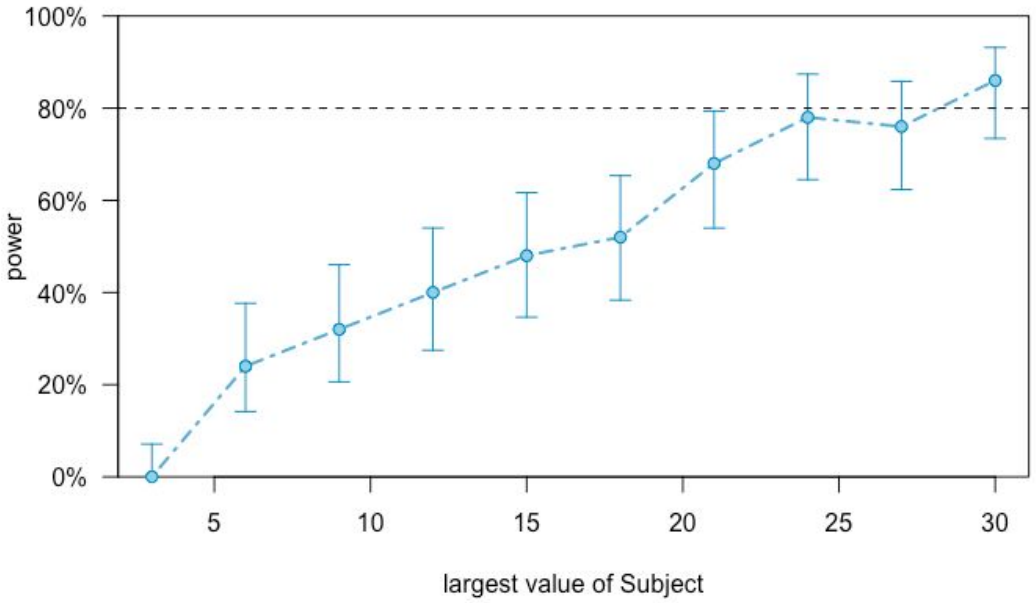

Figure 4: Power curve for the dataset with 30 participants.

Extending visit

(Sigrid) We cannot recruit more participants to gain more power, but we can extend the number of visits to gain more power. Figure 5 shows the result of extending to 10 visits and it shows that 7 visits with 30 participants would give a power of 92.00% with CI = [80.77, 97.78].

However, we know from the previous parts of the assignment that the effect of MLU is best described by a growth curve model rather than a linear model. This means that the effect will stagnate along visits. Model 1 uses the linear method, why extending visit will give an inaccurate estimate of power for the true effect.

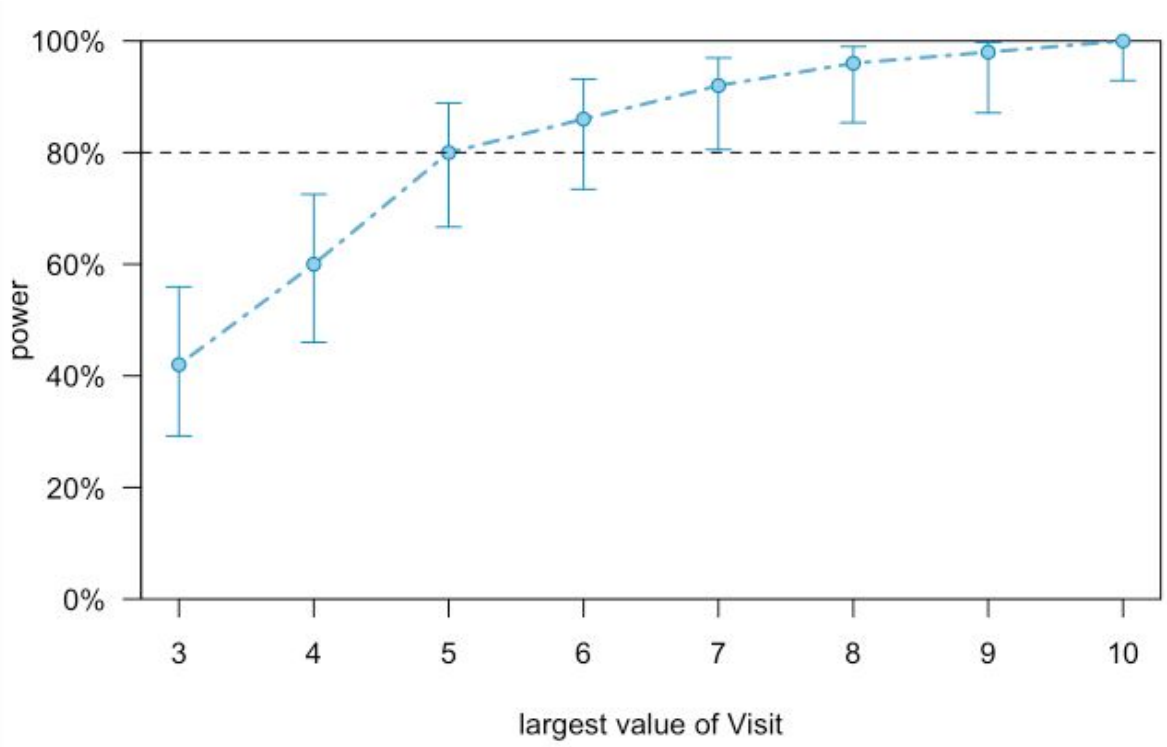

Figure 5: Power curve for an effect size of 0.167 simulating dataset A on model 1 with only 30 participants. Extending visit to 7 is sufficient to get a power of >80% with only 30 participants.

References

- Bates, Maechler, Bolker & Walker (2015). Fitting Linear Mixed-Effects Models Using lme4. Journal of Statistical Software, 67(1), 1-48. doi:10.18637/jss.v067.i01.

- Green & MacLeod (2016). simr: an R package for power analysis of generalised linear mixed models by simulation. Methods in Ecology and Evolution, 7(4), 493-498. doi: 10.1111/2041-210X.12504 (URL: https://doi.org/10.1111/2041-210X.12504), <URL: https://CRAN.R-project.org/package=simr>.